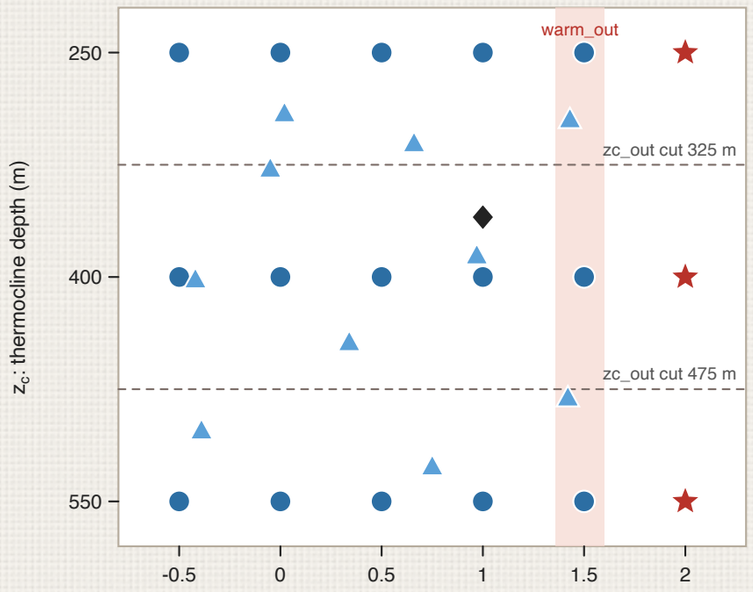
Experiment 3: can a melt emulator trained on 25 cavity runs extrapolate?

1 The runs: 29 MITgcm simulations (Exp. 2)



- 15 factorial runs, s = 1 (training)
- 10 geometry runs, s = 0.8, 1.2 (training)
- 3 final-test runs, T_{bot} = 2.0 (held out)
- Ocean0 reference, linear profile (profile test)

s scales the ice draft; 4 km ISOMIP+ cavity; melt = mean of months 19–24

2 What each method is tested on

Each fit sees only its training runs; standardisation, EOFs, kernels and K, n are refitted inside each fold.

loro — 25 folds: train 24 / test 1
in range: leave one run out

zc_out — 3 folds: thirds of z_c
train 17/16/17, test 8/9/8

warm_out — 1 fold: train $T_{bot} \le 1.0$ (20)
test $T_{bot} \ge 1.4$ (5)

profile_test — train 25 / test Ocean0
different profile; per-cell methods

final — train 25 / test T_{bot} = 2.0 (3)
evaluated once, frozen config

training runs held-out runs

Sensitivity repeat: all folds but final without the 5 runs failing N1 (never used for the hypotheses).

3 Eight ways to predict the melt map

Baseline

M0 mean of the training maps

Physics: one fitted constant

M1 $\gamma\, c\, TF$ — Favier et al. 2019, Eq. 2

M2 $K\, c^2\, |TF|\, TF$ — Burgard et al. 2022, Eq. 14

M3 $K\, c^2\, TF\, |\langle TF \rangle|$ — Jourdain et al. 2020, Eq. 1

Statistics on (T_{bot}, z_c, s)

M4 per-cell regression, degree 1 or 2

M5 Gaussian process on EOFs — Myren & Lawrence 2021

Neural networks, 5 members

M6 deep ensemble on 10 cell features — Lakshminarayanan et al. 2017

M7 hybrid: $K\, TF_+^n\, g_\theta$(geometry) — g_θ sees no temperature

$c = \rho_{sw} c_p / (\rho_i L_i)$; TF: thermal forcing at the ice base

4 How the predictions are scored (fixed before any fit on real data)

Metrics, per held-out run (Burgard et al. 2022):

- RMSE of the melt map over the run's own ice shelf (Eq. 32)
- relative error of the integrated melt (Eq. 33–34)
- error of the mean melt near the grounding line (Eq. 35)
- M6, M7: spread of the 5 network members; M5–M7: coverage, CRPS
- noise floor: no difference below 0.157 m a^{-1} or 1.3 % is read

Pre-registered hypotheses:

H3a On the 3 final-test runs, the hybrid M7 has a smaller integrated-melt error than M6, on average and in at least 2 of the 3 runs (a difference below 1.3 % is inconclusive).

H3b Inside the training range (loro), the hybrid costs little: RMSE of M7 ≤ 1.10 × that of M6.

H3c Out of range (zc_out, warm_out, final: 33 cases), the ensemble is never confidently wrong: no case with spread < 5 % and integrated error > 20 % (Ockenden et al. 2026 replication).

The ranking of all 8 methods is reported as found, whichever wins.

A schematic of the protocol: no model output or result is drawn in this figure.